

Parent Guide



@twinklparents

We're excited to share this activity with you. If you are interested in finding more engaging, fun and interesting activities for you and your children, then check out these links to different areas of the [Twinkl Parents](#) website.

games



crafts



puzzles



experiments



word searches



What is this resource and how do I use it?

These mats cover a variety of strategies for solving division problems, from key stage 1 skills such as sharing, using arrays, number lines and repeated subtraction, to key stage 2 strategies of short and long division, both with and without remainders.

What skills does this practise?

Short Division

Long Division

Remainders

Further Activity Ideas and Suggestions

These [Place Value Charts](#) are another great resource to add to your child's toolkit – pair them with these [Place Value Counters](#) or these [Base Ten Block Cut-Outs](#) to help your child to visualise large numbers.

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Division Strategy Mat 1: Sharing, Repeated Subtraction, Arrays and Number Lines

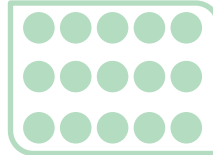
Sharing

Share 15 counters equally between 5 circles.



Arrays

Put your 15 counters into rows of 5.



How many rows did you make?
3 rows

Repeated Subtraction

Subtract 5 from 15 until you reach zero.

$$15 - 5 = 10$$

$$10 - 5 = 5$$

$$5 - 5 = 0$$

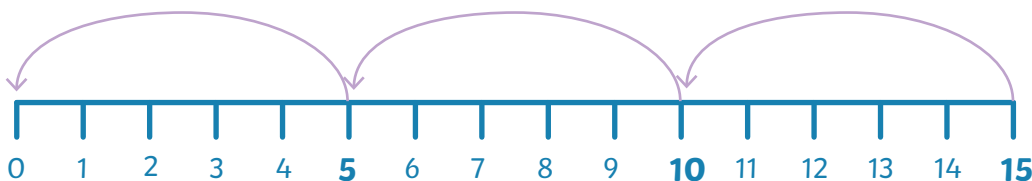
How many times did you subtract 5?

3 times

$$15 \div 5 = 3$$

Number Line

Start at 15. Count backwards in steps of 5.



How many jumps of 5 did it take for you to reach 0?

3 jumps

Division Strategy Mat 2: Short Division (2-digit and 3-digit numbers $\div$ 1-digit numbers)

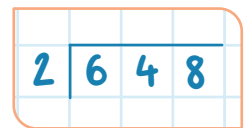
What is the short method?

The short method of division is sometimes also known as the bus stop method. It's a great way of dividing larger numbers quickly.

Let's say we are trying to solve $648 \div 2$.

We can lay out our calculation like this.

The number we are dividing (the dividend) goes inside the 'bus stop' while the number we are dividing it by (the divisor) stays outside.

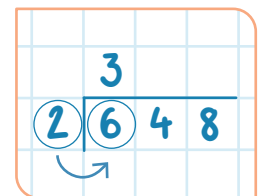


First, solve 6 hundreds $\div$ 2.

We need to put the 6 hundreds into 2 groups.

In other words, count up in 2s until you reach 6 hundreds.

$$600 \div 2 = 300 \text{ (or 3 hundreds)}$$



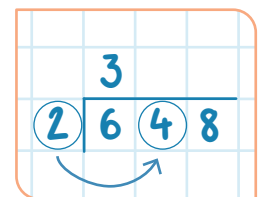
Write your answer – 3 hundreds – above the 6 hundreds, on top of the bus stop.

Next, solve 4 tens $\div$ 2.

We need to put the 4 tens into 2 groups.

In other words, count up in 2s until you reach 4 tens.

$$40 \div 2 = 20 \text{ (or 2 tens)}$$



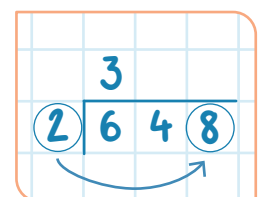
Write your answer – 2 tens – above the 4 tens, on top of the bus stop.

Finally, solve 8 $\div$ 2.

We need to put the 8 ones into 2 groups.

In other words, count up in 2s until you reach 8.

$$8 \div 2 = 4$$



Write your answer – 4 ones – above the 8 ones, on top of the bus stop.

We can now see that $648 \div 2 = 324$



Here is a reminder of what to do if you need to exchange a remainder.

We're going to solve $57 \div 3$.

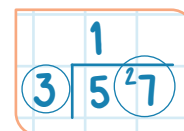
Arrange your numbers like this.



Like before, we need to solve 5 tens $\div$ 3 first.



Put the 5 tens into 3 groups. There would be 1 ten in each group, with 2 tens left over.



Place the 1 ten above the 5 tens, then the 2 leftover tens are exchanged into ones to go into the next column.

Now, we have 27 ones in the next column.

Solve $27 \div 3$.

Put 27 into 3 groups. How many times can you count up in 3s until you reach 27?

The answer is 9. Place the 9 above the 7 ones.

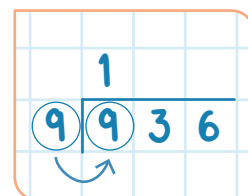
Therefore, $57 \div 3 = 19$.

Now that we are working with larger numbers, we might come across a situation where we need to exchange digits within the calculation.

Let's solve $936 \div 9$.

The first step is straightforward: 9 hundreds divided by 9.

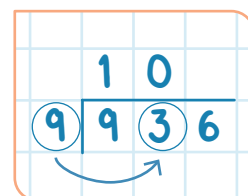
$900 \div 9 = 100$ (or 1 hundred)



The second step is a little trickier.

3 tens divided by 9 can't be done, so we need to write 0 above.

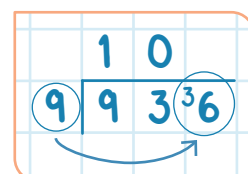
Note: if this is the first digit in your problem, you do not need to write a zero.



Since we couldn't complete that last step, we need to exchange the 3 tens into 30 ones and place them in the next column.

Now we are trying to solve 36 ones divided by 9.

The answer is 4.



So we know that $936 \div 9 = 104$.

Remember:

1. Place the large number (the dividend) inside the bus stop and the small number (the divisor) outside.
2. Divide one number at a time.
3. Exchange any remainders.



Division Strategy Mat 3: Short Division

3 digits $\div$ 1 digit (with remainders)

The short method of division is sometimes also known as the bus stop method.

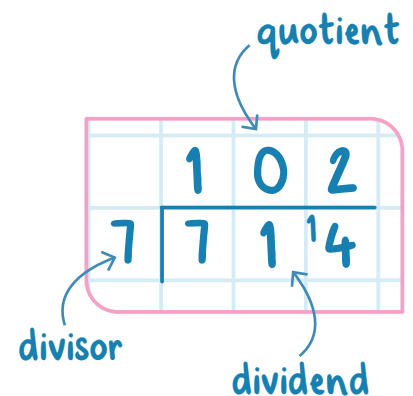
It's a great way of dividing larger numbers quickly.

In a division calculation, we name the numbers according to their role in the calculation:

The number we are dividing by is called the divisor.

The number that is being divided is called the dividend.

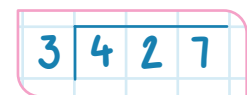
The answer is called the quotient.



Let's say we are trying to solve $427 \div 3$.

We can lay out our calculations like this.

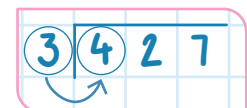
The dividend goes inside the 'bus stop' while the divisor stays outside.



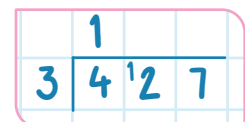
The first step is to divide 4 hundreds between 3.

In other words, count up in steps of 3 until you get to 4.

We can count one lot of 3, and then there is a remainder of 1.



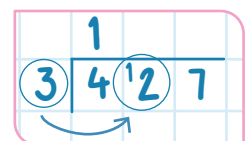
We can write the 1 above the hundreds column and exchange the remaining 1 for tens, in the tens column. This gives us 12 tens.



Now we divide 12 tens between 3.

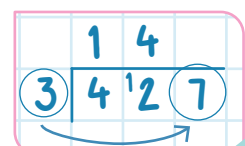
Count up in steps of 3 until you reach 12.

$$12 \div 3 = 4$$



Once we have written the 4 above the tens column, we can divide the 7 ones between 3.

We can count up in steps of 3 until we reach 7. The answer is 2 with a remainder of 1.



We don't have any other numbers to exchange, so we include the remainder in our answer. Instead of writing remainder, we can simply write r.

	1	4	2	r1
3	4	2	7	

We now have the answer: $427 \div 3 = 142r1$

Remember:

1. Place the dividend inside the bus stop and the divisor outside.
2. Divide one number at a time.
3. Write any remainder in the answer with an r.



Division Strategy Mat 4: Short Division

3 digits $\div$ 2 digits

Let's say we are trying to solve $432 \div 12$.

We can lay out our calculation like this.

The dividend goes inside the 'bus stop' while the divisor stays outside.

When we are dividing by 2-digit divisors, it can be useful to jot down several multiples of the divisor to help you. Here, we have written out the first ten multiples of 12 to get us started.

1	2	4	3	2
---	---	---	---	---

Multiples of 12

12 24 36 48 60 72 84 96 108 120

The first step is to divide 4 hundreds between 12.

In other words, count up in steps of 12 until you get to 4.

This can't be done, so we need to exchange our 4 hundreds for 40 tens. We now have 43 tens.

1	2	4	3	2
---	---	---	---	---

Now, we can divide 43 tens between 12.

Count up in steps of 12 until you reach 43.

We can get as far as 36, which is 3 steps of 12.

There is a remainder of 7.

1	2	4	3	2
---	---	---	---	---

Write the 3 above the tens column. The remainder of 7 tens is exchanged for 70 ones. We now have 72 ones.

1	2	4	3	2
---	---	---	---	---

Now, divide 72 ones by 12.

Count up in steps of 12 until you get to 72.

$$72 \div 12 = 6$$

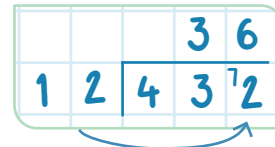
Write 6 above the ones column.

1	2	4	3	2
---	---	---	---	---

We have our answer:

$$432 \div 12 = 36$$

			3	6
1	2	4	3	2



Remember:

1. Place the dividend inside the bus stop and the divisor outside.
2. Write down the multiples of the divisor.
3. Divide each digit of the dividend by the divisor, as you would with 1-digit division.





Division Strategy Mat 5:



Short Division 3 digits $\div$ 2 digits (with remainders)

Let's say we are trying to solve $647 \div 15$.

1	5	6	4	7
---	---	---	---	---

We can lay out our calculations like this.

The dividend goes inside the 'bus stop' while the divisor stays outside.

When we are dividing by 2-digit divisors, it can be useful to jot down several multiples of the divisor to help you. Here, we have written out the first ten multiples of 15 to get us started.

Multiples of 15

15 30 45 60 75 90 105 120 135 150

The first step is to divide 6 hundreds between 15.

In other words, count up in steps of 15 until you get to 6.

1	5	6	4	7
---	---	---	---	---

We can't do that, so we need to combine the hundreds with the tens.

That makes 64 tens.

1	5	6	4	7
---	---	---	---	---

Now, we can count up in steps of 15 until we reach 64.

Looking at our multiples of 15, we can see that we can count 4 steps of 15 to get to 60. There is a remainder of 4.

Write the 4 tens above the tens column and exchange the remaining 4 tens into ones. Place them in the ones column to give us 47 ones.

			4	
1	5	6	4	7

Now, divide 47 ones by 15.

Count up in steps of 15 until you get to 47.

Looking at our multiples of 15, we can see that we can count 3 steps of 15 to get to 45. There is a remainder of 2.

			4	
1	5	6	4	7

Write the 3 above the ones column.

We don't have any other numbers to divide, so we include the remainder in our answer. Instead of writing remainder, we can simply write r.

			4	3	r2
1	5	6	4	7	

We now have the answer: $647 \div 15 = 43r2$

Remember:

Place the dividend inside the bus stop and the divisor outside.

Divide one number at a time.

Exchange any remainders and use them in the next column.

Write any remainder in the final answer with an r.



Division Strategy Mat 6:

Long Division 3 digits $\div$ 2 digits

What is the long method?

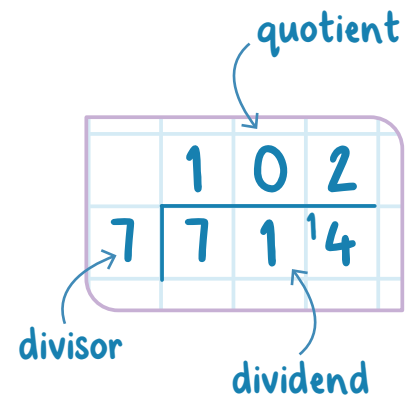
The long method of division is useful for breaking down large division problems into smaller, more manageable chunks. You will need strong times tables knowledge and great subtraction skills to master this method, as well as a lot of space on your paper!

In a division calculation, we name the numbers according to their role in the calculation:

The number we are dividing by is called the **divisor**.

The number that is being divided is called the **dividend**.

The answer is called the **quotient**.



Let's say we are solving $684 \div 12$.

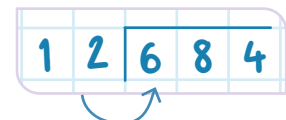
When we are dividing by 2-digit divisors, it can be useful to jot down several multiples of the divisor to help you. Here, we have written out the first ten multiples of 12 to get us started.

Multiples of 12

12 24 36 48 60 72 84 96 108 120

Long division is laid out similarly to short division. The difference is that we will need more space below our problem, which is where we will be doing our 'chunking'.

First, divide the hundreds. 6 hundreds divided by 12 equals 0 hundreds with 6 hundreds remaining. This means we need to exchange. 6 hundreds is the same as 60 tens. The tens column has 8 tens already. Instead of 6 hundreds and 8 tens, think of it as 68 tens.



Next, divide the tens. If we look at our multiples of 12, we can see that 68 tens divided by 12 equals 5 tens because 5 lots of 12 equals 60. We have a remainder of 8 tens.

Write the 5 in the tens column of the answer, above the line. Then, subtract the 60 tens we have shared equally from the 68 tens we started with. This gives us 8 tens remaining.

			5	
1	2	6	8	4
	-	6	0	
			8	

The remaining 8 tens are exchanged for 80 ones and are added to the 4 ones from 684, giving us 84 ones.

			5	
1	2	6	8	4
	-	6	0	↓
			8	4

Finally, divide the ones. We can see that 84 ones divided by 12 equals 7 ones, because 7 lots of 12 equals 84.

When we subtract the 84 ones that we have divided from the 84 ones that we had, we have no remainder.

Therefore, $684 \div 12 = 57$.

			5	7
1	2	6	8	4
	-	6	0	↓
			8	4
		-	8	4
				0

Division Strategy Mat 7: Long Division

4 digits $\div$ 2-digits (with remainders)

Let's say we are solving $6841 \div 14$.

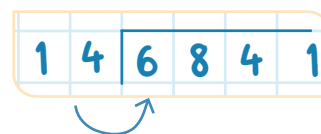
When we are dividing by 2-digit divisors, it can be useful to jot down several multiples of the divisor to help you. Here, we have written out the first ten multiples of 14 to get us started.

Multiples of 14

14 28 42 56 70 84 98 112 126 140

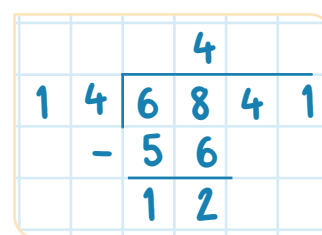
Long division is laid out similarly to short division. The difference is that we will need more space below our problem, which is where we will be doing our 'chunking'.

First, divide the thousands. 6 thousands divided by 14 equals 0 thousands with 6 thousands remaining. This means we need to exchange. 6 thousands is the same as 60 hundreds. The hundreds column has 8 hundreds already. Instead of 6 thousands and 8 hundreds, think of it as 68 hundreds.

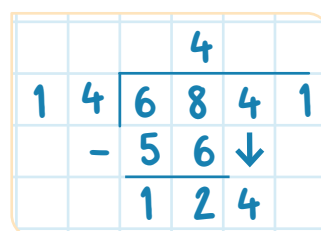


Next, divide the hundreds. If we look at our multiples of 14, we can see that 68 hundreds divided by 14 equals 4 hundreds because 4 lots of 14 equals 56. We have a remainder of 12 hundreds.

Write the 4 in the hundreds column of the answer, above the line. Then, subtract the 56 hundreds we have shared equally from the 68 hundreds we started with. This gives us 12 hundreds remaining.



The remaining 12 hundreds are exchanged for 120 tens and are added to the 4 tens from 6841, giving us 124 tens.



Now, divide the tens. We can see that 124 tens divided by 14 equals 8 tens, because 8 lots of 14 equals 112 with a remainder of 12. Write the 8 tens in the tens column of the answer.

			4	8	
1	4	6	8	4	1
	-	5	6	↓	
		1	2	4	
		1	1	2	
			1	2	

The remaining 12 tens are exchanged for 12 ones and added to the 1 one we have in 6841, giving us 121 ones.

			4	8	
1	4	6	8	4	1
	-	5	6	↓	
		1	2	4	
		1	1	2	↓
			1	2	1

Finally, divide the ones.

We can see that 121 divided by 14 is 8 with a remainder of 9. We can write the 8 in the ones column of the answer.

When we subtract the 112 ones that we have divided from the 121 ones that we had, we have a remainder of 9.

Therefore, $6841 \div 14 = 488\text{r}9$.

			4	8	8	r9
1	4	6	8	4	1	
	-	5	6	↓		
		1	2	4		
		1	1	2	↓	
			1	2	1	
		-	1	1	2	
					9	